



## **panglm: Generalized Linear Models for Panel Data in R**

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### **Abstract**

Panel data combine repeated observations with outcome distributions that may be continuous, binary, or discrete. Existing software often separates linear panel estimators, generalized linear models, high-dimensional fixed effects, and mixed models across different interfaces and likelihood conventions. The R package **panglm** provides a unified formula interface for pooled, fixed-effects, and random-effects generalized linear models with Gaussian, Poisson, binomial, and negative binomial responses. Its compiled numerical core implements iteratively reweighted least squares, exact conditional likelihoods, absorbed one-way and two-way effects, closed-form mixture likelihoods, and Gauss-Hermite quadrature. The package supplies model-based and sandwich covariance estimators, prediction methods with explicit conditional or marginal semantics, diagnostics, hurdle models, and standard model-object methods. This paper defines each statistical model, describes the software architecture, and demonstrates reproducible comparisons with established implementations. The discussion emphasizes where apparently similar panel count models have different likelihoods and interpretations.

*Keywords:* panel data, generalized linear models, fixed effects, random effects, conditional likelihood, count data, R.

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## **1. Introduction**

Panel and longitudinal data arise when units are observed repeatedly. The resulting dependence can be represented through unit-specific parameters, latent random effects, or covariance estimators that remain valid under within-unit correlation. For a Gaussian response, these choices are supported by mature linear panel software such as **plm** (Croissant and Millo 2008). For non-Gaussian responses, the software landscape is more fragmented. **pglm** provides several likelihood-based panel models (Croissant 2021); **fixest** absorbs high-dimensional fixed effects (Bergé 2018); **survival** supplies exact conditional logistic regression (Therneau 2024);

and **glmmTMB** fits a broad class of generalized linear mixed models (Brooks, Kristensen, van Benthem, Magnusson, Berg, Nielsen, Skaug, Mæchler, and Bolker 2017).

The **panglm** package (Badisy 2026) brings a defined subset of these models under one interface. The objective is not to label all estimators as interchangeable. Instead, the package associates every combination of **model**, **family**, and **effect** with a specific likelihood or estimating procedure. This distinction matters particularly for negative binomial panel models, where conditional, unconditional dummy-variable, and mixture likelihoods have different parameter interpretations.

The package is implemented in R (R Core Team 2026) with a numerical core in C++, using **Rcpp** (Eddelbuettel and François 2011), **RcppArmadillo** (Eddelbuettel and Sanderson 2014), and **RcppParallel** (RStudio, PBC 2025). This paper first defines the supported estimators, then describes implementation and inference, illustrates the interface, and summarizes computational validation.

## 2. Statistical models

Let  $y_{it}$  denote the response for individual  $i = 1, \dots, N$  at occasion  $t = 1, \dots, T_i$ , and let  $x_{it}$  be a  $k$ -vector of observed covariates. A general linear predictor is

$$g(\mu_{it}) = x_{it}^\top \beta + \alpha_i + \gamma_t, \quad (1)$$

where  $\alpha_i$  is an individual effect and  $\gamma_t$  is an optional time effect. The treatment of  $\alpha_i$  separates pooled, fixed-effects, and random-effects estimators.

### 2.1. Pooled models

Pooled Gaussian, Poisson, and binomial models omit  $\alpha_i$  and  $\gamma_t$  from Equation 1. They are fitted by iteratively reweighted least squares. The pooled negative binomial model uses the NB2 variance function

$$\text{Var}(y_{it} \mid x_{it}) = \mu_{it} + \mu_{it}^2 / \theta \quad (2)$$

and alternates coefficient updates with estimation of the positive shape parameter  $\theta$ . Pooled point estimates ignore the panel structure, so cluster-robust covariance estimation is often appropriate.

### 2.2. One-way fixed effects

For Gaussian responses, the individual effect is removed by the within transformation. With  $\bar{y}_i = T_i^{-1} \sum_t y_{it}$  and  $\bar{x}_i = T_i^{-1} \sum_t x_{it}$ , least squares is applied to  $y_{it} - \bar{y}_i$  and  $x_{it} - \bar{x}_i$ . Time-invariant regressors are not identified.

For Poisson responses, let  $Y_i = \sum_t y_{it}$  and  $L_i(\beta) = \sum_t \exp(x_{it}^\top \beta)$ . The individual intercept has the profile solution

$$\hat{\alpha}_i(\beta) = \log\{Y_i / L_i(\beta)\}. \quad (3)$$

Substitution yields the conditional Poisson likelihood of Hausman, Hall, and Griliches (1984).

For binomial responses, **panglm** implements the exact conditional logistic likelihood of Chamberlain (1980). If  $S_i = \sum_t y_{it}$ , the contribution of stratum  $i$  is

$$L_i(\beta \mid S_i) = \frac{\exp(\sum_t y_{it} x_{it}^\top \beta)}{\sum_{a: \sum_t a_t = S_i} \exp(\sum_t a_t x_{it}^\top \beta)}. \quad (4)$$

The denominator and inclusion probabilities are computed by a forward and backward dynamic program. All-zero and all-one strata contain no information for  $\beta$ .

The one-way fixed-effects NB2 estimator jointly estimates  $\beta$ , one intercept per individual, and  $\theta$ . This is the unconditional dummy-variable specification discussed by Allison and Waterman (2002), not the conditional negative binomial model. Explicit intercepts avoid interpreting the scale restriction of the latter as fixed effects, but they do not eliminate the general finite- $T$  incidental-parameter concern for a nonlinear fixed-effects estimator.

### 2.3. Two-way fixed effects

Gaussian two-way effects are removed by alternating projections (Guimarães and Portugal 2010). Poisson and NB2 models use an outer iteratively reweighted least-squares loop and inner weighted projections. The algorithm updates  $\beta$  using variables partialled out with respect to individual and time effects, then reconstructs the complete linear predictor. A general exact two-way conditional logistic likelihood is not available, so the binomial fixed-effects implementation is one-way.

### 2.4. Random effects

The Gaussian random-effects estimator uses Swamy-Arora feasible generalized least squares (Swamy and Arora 1972). The random-effects Poisson model integrates a mean-one Gamma frailty, producing a closed-form Poisson-Gamma marginal likelihood. The random-effects negative binomial model uses a distinct beta-negative-binomial likelihood with two positive shape parameters. For binomial responses, a Gaussian random intercept is integrated by Gauss-Hermite quadrature. Response predictions from random-effects models are marginal over the fitted mixing distribution.

### 2.5. Hurdle counts

The function **panglm\_hurdle()** separates the occurrence and intensity processes for panel counts. The first component is an exact conditional logistic model for  $I(y_{it} > 0)$ . The second component is a zero-truncated Poisson or NB2 model for positive observations, with density

$$f_+(y \mid \vartheta) = \frac{f(y \mid \vartheta)}{1 - f(0 \mid \vartheta)}, \quad y > 0. \quad (5)$$

This follows the standard hurdle decomposition of Mullahy (1986) while including individual dummy intercepts in the positive-count component.

## 3. Inference and model methods

Model-based covariance matrices are obtained from the relevant least-squares system or observed information. For pooled and fixed-effects estimators, **panglm** also supplies HC1 and cluster-robust sandwich covariances. With score contributions  $s_j(\hat{\beta})$  and inverse information  $B$ , the generic form is

$$\hat{V}(\hat{\beta}) = B \left\{ \sum_j s_j(\hat{\beta}) s_j(\hat{\beta})^\top \right\} B. \quad (6)$$

For conditional logistic regression, the independent likelihood unit is a complete stratum. Observation components are therefore summed within strata before the meat of Equation 6 is formed. A custom cluster may combine complete strata but cannot split one. Small-sample corrections follow the HC1 and cluster conventions described by [Cameron and Miller \(2015\)](#).

Likelihood metadata count shape, variance, dispersion, and explicit fixed effects when they belong to the reported likelihood. Conditional Poisson and conditional logistic likelihoods count only parameters remaining after conditioning. Consequently, `AIC()` and `BIC()` are available for likelihood-based fits but not for Gaussian within or Swamy-Arora estimators.

Objects of class ‘**panglm**’ implement methods for `coef()`, `summary()`, `vcov()`, `confint()`, `logLik()`, `fitted()`, `residuals()`, `predict()`, and `plot()`. Methods for `tidy()` and `glance()` support table-building workflows. In-sample values preserve the row order supplied by the user. Fixed-effects new-data predictions require known individual and time levels, while exact conditional binomial fits only provide probabilities conditional on an observed stratum total.

## 4. Software design

The primary interface is

```
panglm(formula, data, index, model, family, weights, offset,
       na.action, effect, vcov, R, maxit, tol)
```

The `index` argument gives the individual and optional time columns. The formula machinery supports factors and interactions. The model frame is aligned with the panel indexes, then sorted into contiguous individual blocks for the compiled routines. Returned fitted values and residuals are mapped back to input order.

Before fitting, rank is checked after the transformation appropriate to the estimator. This detects ordinary collinearity as well as regressors removed by one-way or two-way fixed effects. Unidentified columns are reported and omitted. Missing responses and covariates follow `na.action`; missing or duplicate panel keys are errors. Nontrivial weights and nonzero offsets are rejected because weighted conditional likelihoods require statistical models that are not implemented by the current routines.

The compiled layer parallelizes independent reductions over panel units. It contains kernels for grouped aggregation, demeaning, likelihood evaluation, scores, Hessians, and quadrature. Optimization is performed by Newton or quasi-Newton updates with step control. Scaling is applied internally to reduce conditioning problems and coefficients are transformed back before the fitted object is assembled.

| Model  | Family                      | Estimator                    | Reference comparison |
|--------|-----------------------------|------------------------------|----------------------|
| Pooled | Gaussian, Poisson, binomial | IRLS                         | <b>glm()</b>         |
| Pooled | NB2                         | Profiled IRLS                | <b>glm.nb()</b>      |
| Within | Gaussian                    | Demeaned least squares       | <b>plm()</b>         |
| Within | Poisson                     | Conditional likelihood       | <b>pglm, fixest</b>  |
| Within | Binomial                    | Exact conditional likelihood | <b>clogit()</b>      |
| Within | NB2                         | Dummy-variable likelihood    | <b>fixest</b>        |
| Random | Gaussian                    | Swamy-Arora FGLS             | <b>plm()</b>         |
| Random | Poisson, NB, binomial       | Marginal likelihood          | <b>pglm</b>          |
| Hurdle | Poisson, NB2                | Truncated likelihood         | <b>pscl</b>          |

Table 1: Estimator families and principal reference implementations used in the package test suite. A listed comparison indicates agreement only where the statistical likelihoods coincide.

## 5. Illustrations

### 5.1. Gaussian investment panel

The `Grunfeld` data contain annual observations for ten firms. Pooled, one-way fixed-effects, and random-effects models can be fitted with the same formula.

```
R> data("Grunfeld", package = "plm")
R> pool_fit <- panglm(inv ~ value + capital, data = Grunfeld,
+   index = c("firm", "year"), model = "pooling",
+   family = "gaussian")
R> within_fit <- panglm(inv ~ value + capital, data = Grunfeld,
+   index = c("firm", "year"), model = "within",
+   family = "gaussian")
R> random_fit <- panglm(inv ~ value + capital, data = Grunfeld,
+   index = c("firm", "year"), model = "random",
+   family = "gaussian")
R> cbind(pooling = coef(pool_fit), random = coef(random_fit))
```

|             | pooling     | random      |
|-------------|-------------|-------------|
| (Intercept) | -42.7143694 | -57.8344149 |
| value       | 0.1155622   | 0.1097812   |
| capital     | 0.2306785   | 0.3081130   |

The within estimates agree with **plm** up to numerical precision.

```
R> plm_fit <- plm(inv ~ value + capital, data = Grunfeld,
+   index = c("firm", "year"), model = "within")
R> rbind(panglm = coef(within_fit), plm = coef(plm_fit))
```

```

      value    capital
panglm 0.1101238 0.3100653
plm    0.1101238 0.3100653

```

```
R> max(abs(coef(within_fit) - coef(plm_fit)))
```

```
[1] 3.885781e-16
```

The pooled coefficient estimates can be paired with a covariance clustered by firm without refitting the model.

```
R> sqrt(diag(vcov(pool_fit, type = "cluster")))
```

```

(Intercept)      value      capital
20.42520293  0.01589434  0.08496711

```

```
R> confint(pool_fit)
```

```

              2.5 %      97.5 %
(Intercept) -61.3569119 -24.0718270
value        0.1041244   0.1269999
capital      0.1807468   0.2806101

```

## 5.2. Panel count regression

A reproducible Poisson panel is generated from the covariates in **Grunfeld**. The individual fixed-effects estimator conditions on each firm's total count.

```

R> set.seed(42)
R> Grunfeld$count <- rpois(nrow(Grunfeld),
+   lambda = exp(0.5 + 0.0002 * Grunfeld$value))
R> poisson_fe <- panglm(count ~ value + capital, data = Grunfeld,
+   index = c("firm", "year"), model = "within",
+   family = "poisson")
R> summary(poisson_fe)

```

Call:

```
panglm(formula = count ~ value + capital, data = Grunfeld, index = c("firm",
"year"), model = "within", family = "poisson")
```

```

Family: poisson  Link: log  Model: within  Effect: individual
N = 200  groups = 10  vcov: classical

```

Coefficients:

```
Estimate Std. Error z value Pr(>|z|)
```

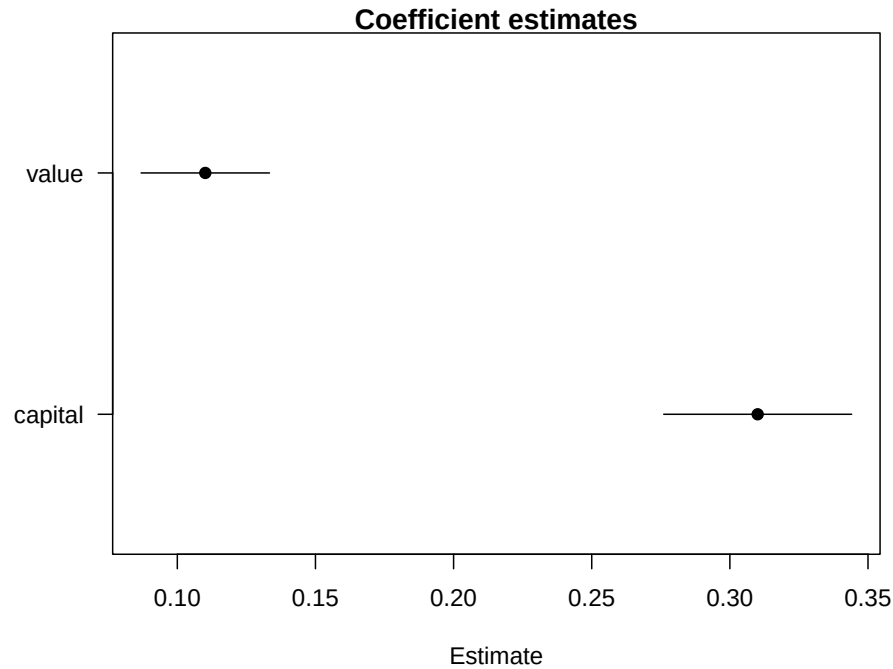


Figure 1: Coefficient estimates and 95 percent Wald intervals from the Gaussian fixed-effects model.

```
value    1.012e-04  1.152e-04  0.879  0.380
capital -4.581e-05  1.653e-04 -0.277  0.782
```

```
Log-likelihood: -311.8
Dispersion parameter: 1
Residual df: 188  Iterations: 3
```

### 5.3. Model visualization

The plot method displays coefficient intervals for any fitted model and diagnostic views when fitted values are defined. Figure 1 shows the within-estimator coefficients.

## 6. Validation and scope

The test suite uses equality comparisons when another implementation fits the same statistical model. It covers pooled models against **stats** and **MASS** (Venables and Ripley 2002), Gaussian panel models against **plm**, exact conditional logistic regression against **survival**, absorbed count effects against **fixest**, compatible marginal likelihoods against **pglm**, and truncated counts against **pscl** (Jackman, Tahk, Zeileis, Maimone, Fearon, Meers, and Katz 2024). Additional tests cover sparse and unbalanced panels, boundary groups, rank deficiency, prediction semantics, model metadata, and original-row-order preservation.

The package intentionally does not provide between estimators, ordinal or Tobit families, nontrivial observation weights, or nonzero offsets. A two-way exact conditional logistic esti-

mator is also outside its scope. These restrictions are checked at the interface so unsupported inputs cannot produce apparently valid results.

## 7. Summary and discussion

**panglm** provides one interface for a broad but explicitly defined set of panel generalized linear models. The implementation combines classical panel transformations, conditional likelihoods, absorbed effects, mixture models, and numerical quadrature. Standard model methods expose inference, information criteria, prediction, diagnostics, and visualization without hiding differences between conditional and marginal quantities.

Further work could add weighted estimators, offsets, ordinal responses, and other random-effect distributions. These extensions should enter as new statistical implementations with their own validation, not as argument handling layered on an incompatible likelihood.

## Computational details

The results in this paper were obtained using R 4.5.1 and **panglm** 0.6.0. The package is distributed under the MIT license and is available from <https://github.com/ielbadisy/panglm>.

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